SIMATS SCHOOL OF ENGINEERING

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Course Code: DSA0311	Course Name: Natural Language Processing
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Assignment 5

1. Write a finite-state automaton for a dialogue manager for checking your bank balance and withdrawing money at an automated teller machine. (10M) (CO5) (BLT4)

2. Consider the sentence "Christ opened the eyes", describes fact about English syntax. a) Describe and illustrate how your favourite foreign language differs in this respect. b) Explain how a MT system could deal with this difference. (10M) (CO5) (BLT4)

QUESTION 1 – FINITE-STATE AUTOMATON FOR AN ATM DIALOGUE MANAGER (10 MARKS)

1. Introduction

A Finite-State Automaton (FSA) is a mathematical model used to represent a system that moves between different states based on user input. It is widely used in Natural Language Processing, speech recognition and dialogue management.

An Automated Teller Machine (ATM) allows customers to perform banking operations such as checking their account balance and withdrawing money.

A dialogue manager controls the interaction between the customer and the ATM. It understands the customer's request, asks for the required information and performs the selected operation.

2. Objectives

1. To design a finite-state automaton for an ATM system.
2. To identify the different states of ATM operations.
3. To handle balance checking and money withdrawal.
4. To provide suitable responses to customer requests.
5. To complete transactions safely and correctly.

3. What Is an ATM Dialogue Manager?

An ATM dialogue manager is a system that controls the conversation between the user and the ATM.

It follows a sequence of steps to complete a transaction.

Example conversation:

ATM	Customer
Welcome to ATM	—
Enter your PIN	Enters PIN
Select an operation	Check balance
Your balance is ₹10,000	—
Thank you	—

The dialogue manager changes its state according to the customer's response.

4. States of the ATM System

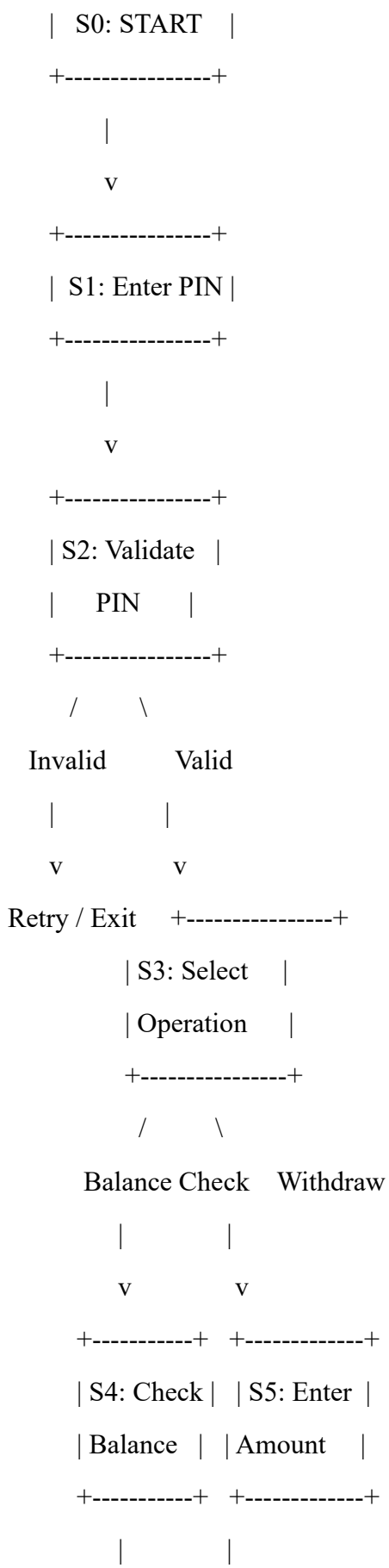
The ATM system contains the following states:

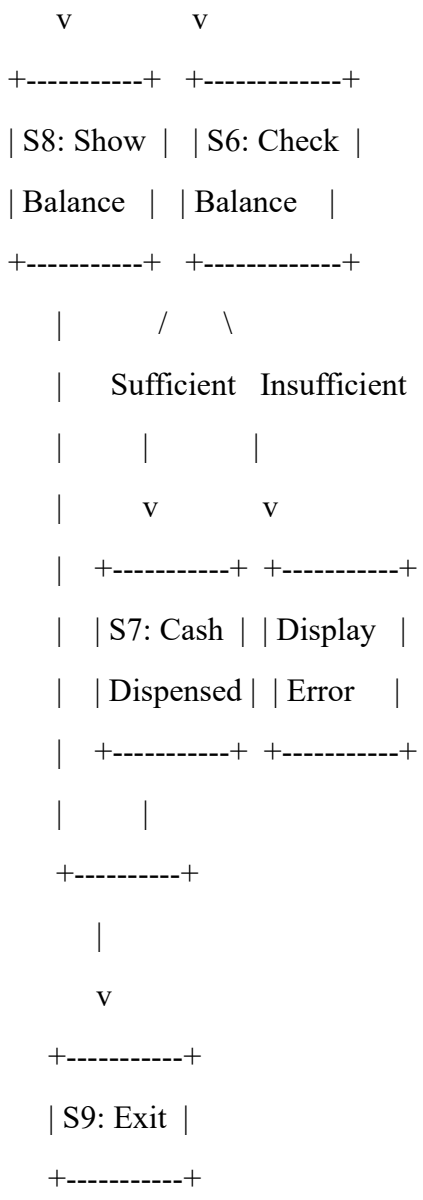
State	Description
S0	Start
S1	Enter PIN
S2	Validate PIN
S3	Select Operation
S4	Check Balance
S5	Enter Withdrawal Amount
S6	Check Available Balance
S7	Dispense Cash
S8	Display Result
S9	Exit

5. State Transition Diagram

Simplified diagram of the ATM dialogue process

```
+----------------+
```


S0: START
S1: Enter PIN
S2: Validate PIN
Invalid
Valid
Retry / Exit
S3: Select Operation
Balance Check
Withdraw
S4: Check Balance
S5: Enter Amount



6. Working of the ATM Dialogue Manager

Step 1: Start

The ATM welcomes the customer and starts the transaction.

Step 2: Enter PIN

The customer enters the personal identification number.

Step 3: Validate PIN

The ATM checks whether the entered PIN is correct. If it is incorrect, the ATM asks the customer to retry or exits after the permitted number of attempts.

Step 4: Select Operation

The customer selects one of the available options:

- Check balance.

- Withdraw money.
- Exit.

Step 5: Check Balance

If the customer selects balance enquiry, the ATM retrieves and displays the available balance.

Step 6: Enter Withdrawal Amount

If the customer selects withdrawal, the ATM asks the customer to enter an amount.

Step 7: Check Available Balance

The ATM checks whether the account contains sufficient money.

Step 8: Dispense Cash

If sufficient money is available and the ATM has cash, it dispenses the requested amount and updates the account balance.

Step 9: Exit

The ATM displays the transaction result and ends the session.

7. State Transition Table

Current State	**Input**	**Next State**
Start	Begin	Enter PIN
Enter PIN	PIN entered	Validate PIN
Validate PIN	Correct PIN	Select Operation
Validate PIN	Incorrect PIN	Retry or Exit
Select Operation	Balance	Check Balance
Select Operation	Withdraw	Enter Amount
Check Balance	Balance retrieved	Display Result
Enter Amount	Amount entered	Check Available Balance
Check Available Balance	Sufficient funds	Dispense Cash
Check Available Balance	Insufficient funds	Display Error
Dispense Cash	Successful	Display Result
Display Result	Exit	End

8. Mathematical Representation of FSA

A Finite-State Automaton can be represented as:

$$ M=(Q,\Sigma,\delta,q_0,F) $$

Where:

- Q = Set of states.
- Σ = Set of input symbols.
- δ = State transition function.
- q_0 = Initial state.
- F = Set of final states.

For the ATM system:

Q:

$$ \{S0,S1,S2,S3,S4,S5,S6,S7,S8,S9\} $$

Σ:

$$\{PIN,VALID,INVALID,BALANCE,WITHDRAW,
AMOUNT,SUFFICIENT,INSUFFICIENT,EXIT\}$$

Initial State:

$$ q_0=S0 $$

Final State:

$$ F=\{S9\} $$

The transition function decides the next state based on the current state and the user's input.

9. Example 1: Checking Bank Balance

Suppose a customer wants to check the bank balance.

Step	Customer Action	ATM Response
1	Starts ATM	Welcome
2	Enters PIN	PIN validated
3	Selects Balance	Retrieving balance
4	—	Balance displayed
5	Exits	Transaction completed

Output:

Welcome to ATM

Enter PIN

PIN Verified

Select Operation

Balance: Rs. 10,000

Thank You

10. Example 2: Withdrawing Money

Suppose a customer wants to withdraw ₹2,000 from an account containing ₹10,000.

Step	Customer Action	ATM Response
1	Starts ATM	Welcome
2	Enters PIN	PIN validated
3	Selects Withdraw	Enter amount
4	Enters ₹2,000	Checking balance
5	—	Sufficient balance
6	—	Cash dispensed
7	—	New balance ₹8,000
8	Exits	Thank you

Output:

Welcome to ATM

Enter PIN

PIN Verified

Select Withdrawal

Enter Amount: 2000

Cash Dispensed

Remaining Balance: 8000

Thank You

11. Python Program

The following simple program demonstrates balance checking and withdrawing money using an ATM.

```
balance = 10000

pin = 1234
```

```
print("Welcome to ATM")
entered_pin = int(input("Enter PIN: "))
if entered_pin == pin:
    print("1. Check Balance")
    print("2. Withdraw Money")
    choice = int(input("Enter choice: "))
    if choice == 1:
        print("Balance:", balance)
    elif choice == 2:
        amount = int(input("Enter amount: "))
        if amount <= balance and amount > 0:
            balance = balance - amount
            print("Cash Dispensed")
            print("Remaining Balance:", balance)
        else:
            print("Insufficient Balance or Invalid Amount")
    else:
        print("Invalid Choice")
else:
    print("Incorrect PIN")
```

OUTPUT:

```
Welcome to ATM
Enter PIN: 1234
1. Check Balance
2. Withdraw Money
Enter choice: 2
Enter amount: 2000
Cash Dispensed
Remaining Balance: 8000
```

12. Advantages of ATM Dialogue Manager

1. Provides easy interaction between users and ATMs.
2. Supports multiple banking operations.
3. Reduces manual work.

4. Provides quick transaction responses.
5. Helps control the transaction sequence.
6. Handles incorrect inputs.
7. Improves customer convenience.

13. Limitations

1. Incorrect PIN entries may block access.
2. Network problems may interrupt transactions.
3. Cash availability can limit withdrawals.
4. A basic FSA may not understand complex natural language requests.
5. Additional states are required for more banking operations.

14. Applications

1. Automated banking systems.
2. Customer service chatbots.
3. Interactive voice response systems.
4. Online banking applications.
5. Self-service payment machines.

15. Conclusion

The ATM dialogue manager uses a Finite-State Automaton to control customer interactions. It moves through different states such as PIN verification, balance checking and money withdrawal. The system provides an organized transaction process and helps customers complete banking operations easily and securely.

QUESTION 2 – FOREIGN LANGUAGE SYNTAX AND MACHINE TRANSLATION

Given sentence: “Christ opened the eyes.”

1. Introduction

Natural Language Processing helps computers understand and process human languages. Every language follows its own grammatical rules and sentence structures.

English generally follows the Subject–Verb–Object (SVO) word order. However, Telugu commonly follows the Subject–Object–Verb (SOV) word order.

This assignment explains the difference between English and Telugu syntax and describes how a Machine Translation (MT) system handles this difference.

2. English Sentence Structure

The given English sentence is:

“Christ opened the eyes.”

It follows the SVO sentence structure.

- Subject: Christ
- Verb: opened
- Object: the eyes

English Word Order

\[Subject+Verb+Object \]

English Syntax Tree

```
          Sentence
         /   |   \
    Subject  Verb  Object
      |      |      |
   Christ opened  the eyes
```

Explanation

1. Christ is the subject performing the action.
2. Opened is the past-tense verb.
3. The eyes is the object receiving the action.
4. The sentence follows Subject–Verb–Object order.

3. Telugu Language Structure

Telugu is a Dravidian language spoken mainly in Andhra Pradesh and Telangana.

Unlike English, Telugu generally follows Subject–Object–Verb (SOV) word order.

The English sentence:

Christ opened the eyes.

A natural Telugu translation is:

క్రీస్తు కళ్ళు తెరిచాడు.

Transliteration:

Kreesthu kallu terichaadu.

Meaning:

Christ opened the eyes.

Telugu Word Order

\[Subject+Object+Verb \]

Telugu Sentence Structure

Telugu word	Meaning	Grammatical role
క్రీస్తు (Kreesthu)	Christ	Subject
కళ్ళు (Kallu)	Eyes	Object
తెరిచాడు (Terichaadu)	Opened	Verb

Telugu Syntax Tree

```
        Sentence
       /   |   \
  Subject  Object  Verb
    |      |      |
  Kreesthu  Kallu  Terichaadu
```

4. Differences Between English and Telugu

Feature	English	Telugu
Word order	SVO	SOV
Subject position	First	First
Object position	After verb	Before verb
Verb position	Middle	Usually last
Example	Christ opened the eyes	Kreesthu kallu terichaadu

Main Difference

English places the verb before the object, whereas Telugu places the object before the verb.

English:

Christ + opened + the eyes.

Telugu:

Kreesthu + kallu + terichaadu.

This is the major syntactic difference between the two languages.

5. Why Do These Differences Occur?

Languages develop different grammatical structures based on their linguistic systems.

1. English generally uses word order to identify grammatical roles.
2. Telugu commonly uses suffixes and grammatical markers to express relationships.
3. English verbs usually appear before objects.
4. Telugu verbs usually appear at the end of sentences.
5. Directly translating word by word may produce incorrect Telugu sentences.

6. What Is Machine Translation?

Machine Translation (MT) is the process of automatically converting text from one language into another using computer systems.

For example:

Source Language: English

“Christ opened the eyes.”

Target Language: Telugu

“క్రీస్తు కళ్ళు తెరిచాడు.”

The MT system analyzes the English sentence, identifies its meaning and generates a grammatically correct Telugu sentence.

7. How Can an MT System Handle This Difference?

A Machine Translation system can handle the difference using the following steps.

Step 1: Input Sentence

The user enters the English sentence:

“Christ opened the eyes.”

Step 2: Tokenization

The sentence is divided into individual words.

Token Number	Word
1	Christ
2	opened
3	the
4	eyes

Step 3: Syntactic Analysis

The system identifies the grammatical roles.

- Christ = Subject
- Opened = Verb
- The eyes = Object

Step 4: Semantic Analysis

The system understands the meaning of the sentence.

It identifies that Christ performed the action of opening the eyes.

Step 5: Word Order Transformation

The system changes the English SVO structure into the Telugu SOV structure.

English Order	Telugu Order
Christ	Christ
Opened	—
The eyes	Eyes
—	Opened

Step 6: Telugu Word Generation

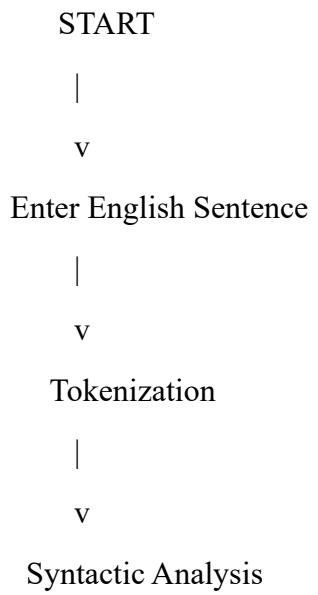
The system selects suitable Telugu words and generates the correct verb form.

Step 7: Output Generation

The final Telugu sentence is generated:

క్రీస్తు కళ్ళు తెరిచాడు.

8. Machine Translation Flowchart



|

v

Identify SVO Structure

|

v

Understand Sentence

|

v

Change SVO into SOV

|

v

Generate Telugu Words

|

v

Check Grammar and Meaning

|

v

Display Telugu Output

|

v

END

9. Example of Word Order Transformation

Stage	**English**	**Telugu**
Subject	Christ	Kreesthu
Verb	Opened	Terichaadu
Object	The eyes	Kallu
Sentence order	SVO	SOV
Final sentence	Christ opened the eyes	Kreesthu kallu terichaadu

The MT system rearranges the sentence according to the grammatical rules of Telugu.

10. Types of Machine Translation

1. Rule-Based Machine Translation

It uses predefined grammar rules and dictionaries to translate sentences.

Example: Changing English SVO order into Telugu SOV order using grammar rules.

2. Statistical Machine Translation

It uses previously translated sentence pairs and statistical patterns to select suitable translations.

3. Neural Machine Translation

It uses neural networks and deep learning to understand sentence meaning and generate translations.

Examples include modern translation systems that translate complete sentences rather than simply replacing individual words.

11. Simple Python Program for Word Order Transformation

The following program demonstrates a basic word-order change. It is only an illustration of the SVO-to-SOV transformation, not a complete English-to-Telugu translation system.

```
sentence = "Christ opened the eyes"

words = sentence.split()

subject = words[0]

verb = words[1]

object = " ".join(words[2:])

print("English:", sentence)

print("Subject:", subject)

print("Verb:", verb)

print("Object:", object)

print("SOV Order:", subject, object, verb)
```

OUTPUT:

```
English: Christ opened the eyes
Subject: Christ
Verb: opened
Object: the eyes
SOV Order: Christ the eyes opened
```

12. Challenges in Machine Translation

1. Different word orders between languages.
2. Differences in grammar rules.

3. Multiple meanings of words.
4. Correct verb tense selection.
5. Cultural and contextual differences.
6. Maintaining the original meaning.

13. Advantages of Machine Translation

1. Helps people communicate in different languages.
2. Saves time in translation.
3. Supports multilingual applications.
4. Helps translate documents and websites.
5. Improves access to information.
6. Supports education and learning.

14. Applications

1. Google Translate and other translation applications.
2. Multilingual chatbots.
3. Website translation.
4. International communication.
5. Educational applications.
6. Automatic document translation.

15. Conclusion

English and Telugu follow different sentence structures. English generally uses Subject–Verb–Object order, whereas Telugu generally uses Subject–Object–Verb order. A Machine Translation system handles this difference by analyzing the sentence structure, understanding its meaning and rearranging the words according to Telugu grammar. This helps generate a natural and meaningful translation.

OVERALL CONCLUSION

This assignment explains the use of Finite-State Automata and Machine Translation in Natural Language Processing. The ATM dialogue manager demonstrates how a system moves between states to perform banking operations. The English and Telugu comparison demonstrates how languages differ in sentence structure and how Machine Translation handles these differences. Both topics show how NLP helps computers interact with users and understand human languages.